Steen 2021 DLBCL — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Whole malignant manifold (2,478 cells, 3 tumours) is one connected component (0 islands). Harmony (θ=10) lifts per-neighbourhood mixing from iLISI 1.00→1.88 of 3 samples; the regenerated embedding shows one continuous cloud with samples intermixed — no per-patient archipelago.

Gate 2 — module geography → GO (non-branching)

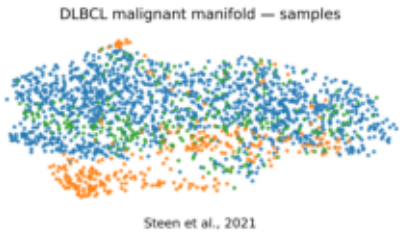
A distinct proliferative lobe (513 cells, 95.5% cycling) confounds Gastrulation on the whole manifold (Gast↔cycling r=−0.74). Scoping to the low-cycling lobe restores the reference relationships: Gast/pSom co-vary (r=+0.6) and both oppose Crest (Crest↔Gast r=−0.9). One (Gast/pSom)→Crest axis — Gast ρ=−0.94, pSom ρ=−0.73 decline; Crest rises to a terminal tip (ρ=+0.87, Crest-high decile pt 0.95; joint pSom&Crest enrichment 0.26<1). Single axis, no bifurcation — GO.

Gate 3 — cross-repression → NA

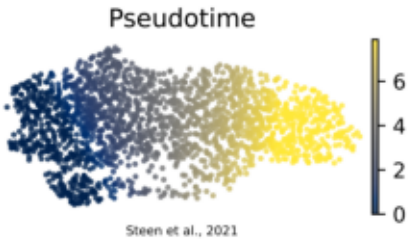
Not applicable: the manifold is non-branching (principal curve, 2 tips / 0 forks, fold-free self-approach 2.9 across Nodes=15/20/30). With no Gast/pSom hub there is no bifurcation to test.

STEP FIGURES

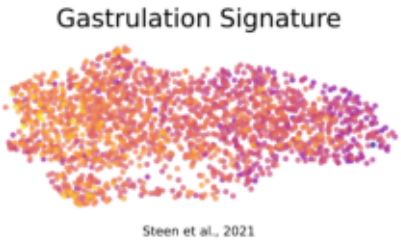
Manifold — samples



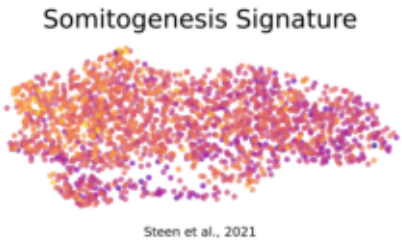
Pseudotime



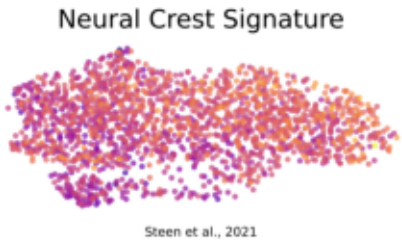
Gastrulation



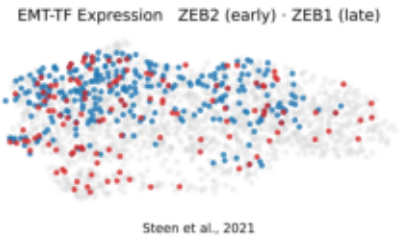
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Diffuse large B-cell lymphoma (Steen et al., 2021), 10x — 2,522 malignant cells → 2,478 after threshold (3 tumours) → 1,963 scoped to the low-cycling developmental lobe. Non-branching Linear-pSom/Crest.

Verdict.
Pattern reproduces (non-branching). Malignant DLBCL cells recapitulate a single gastrulation/somitogenesis → neural-crest developmental axis once the proliferative compartment (513 cells, 95.5% cycling) is scoped out. Gate 1 GO, Gate 2 GO, Gate 3 NA (no bifurcation). EMT-TFs ZEB2 (early/root) → ZEB1 (late/terminus).

Parameter choices

Min-cell threshold	127 (geo-mean of 8.4× cliff 371→44); drops thin DLBCL111B → 3 tumours, 2,478 cells	**Integration**	Harmony on sample θ=10 (mixing floor by θ=5); n_pcs=20, spread=1.3
Scope	low-cycling Leiden lobe (res=0.15): 1,963 cells; excludes 95.5%-cycling lobe	**Geometry / root**	principal curve (ElPiGraph, non-branching); root Gast/pSom-high tip; Linear-pSom/Crest
EMT-TFs	ZEB2 (61%) & ZEB1 (26%) — only factors above 3% floor; rest near-silent in B-cell tumour		